SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

Math 1110: Calculus I - Course Syllabus

1. Basic Course Information

- Course Code and Title: *Math 1101: Calculus I*
- Credit Hours: *4 crs*
- Instructor(s): *Prof. Abdulkadir Hassen, Prof. Aklilu Zeleke, and Dr. Fikreab Solomon*
- Semester/Quarter: *Fall 2025, First Term*
- Prerequisites: *Precalculus*
- Co-requisites: *None*
- Class Schedule: *Monday and Wednesday 2:00pm to 3:15pm and Friday 2:00pm to 4pm.*
- Office Hours: TBA
- Contact Information: *ahassen09@gmail.com* and *zeleke@msu.edu* and *fabmaysolj@gmail.com*

2. Course Description

Differentiation and integration of functions of one variable, with applications. Informal treatment of limits and continuity. Differentiation: definition, rules, application to graphing, rates, approximations, and extremum problems. Indefinite integration; definite integral; fundamental theorem of calculus. Applications of integration to geometry and science. Elementary functions.

3. Course Objectives

Students will demonstrate the ability to:

1. *Identify and apply basic functions*
2. *Compute limits in various forms*
3. *Properties of continuous functions and applications to approximating roots of equations*
4. *Understand the geometric, physical, and algebraic interpretation of derivatives*
5. *Differentiate basic functions using the rules of derivatives*
6. *Use differentiation to solve extreme and related rate problems*
7. *Understand the geometric, physical, and algebraic interpretation of definite integrals*

8. *Explain the connection between derivative and integrals*

9. *Set up integrals to find volumes of regions whose cross-section areas can be found*

10. *Use integration to find areas and volume*

11. *Use integration to find work done by a force and average values of a function*

12. *Solve word problems using derivatives and integration*

13. *Work on group project for deeper understanding of derivative and integration*

4. Student Learning Outcomes (ABET Outcomes 1–7)

Upon successful completion of the course, students will be able to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2. To produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

3. Communicate effectively with a range of audiences.

4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.

5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.

6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

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5. Mapping to Program Outcomes

ABET Outcomes	Course Objectives
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1, 2, 5, 8, 9, 10, 11
2. To produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	3, 7,
3. Communicate effectively with a range of audiences.	7, 9
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.	
5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.	13
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.	2-12

6. Required Textbooks and References

Text: Calculus: Early Transcendentals, 9th Edition, by James Stewart, Daniel K. Clegg, Saleem Watson.

References:

1) *Calculus, Early Transcendental, 4th Edition, by Jon Rogawski, Collin Adams, and Robert Franzosa.*

2) *Calculus, Volume 1: https://openstax.org/details/books/calculus-volume-1*

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7. Course Topics and Schedule

Content: (12 weeks 4 lectures per week = 48 lectures)

Week(s)	Topic	Assignments
1	**Ch 1 Functions and Models** *1.1. Four Ways to Represent a Function* *1.2. Mathematical Models: Essential Functions* *1.3. New Functions from Old Functions* *1.4. Exponential Functions* *1.5. Inverse Functions and Logarithms*	HW (Homework) 1 Q (Quiz) 1
2, 3	**Ch2. Limits and Derivatives** *2.1. The Tangent and Velocity Problems* *2.2. The Limit of a Function* *2.3. Calculating Limits Using the Limit Laws* *2.4. The Precise Definition of a Limit* *2.5. Continuity* *2.6. Limits at Infinity; Horizontal Asymptote* *2.7. Derivatives and Rates of Change* *2.8. The Derivative as a Function*	HW 2, 3, 4 Q 2, 3 Test 1 (Ch 1 and 2)
4, 5, 6	**Ch3. Differentiation Rules** *3.1. Derivatives of Polynomials and Exp. Functions* *3.2. The Product and Quotient Rules* *3.3. Derivatives of Trigonometric Functions* *3.4. The Chain Rule* *3.5. Implicit Differentiation* *3.6. Derivatives of Log and Inverse Trig.* *3.7. Related Rates* *3.8 Linear Approximations and Differentials*	HW 5, 6, 7 Q 4, 5
7 and 8	**Ch4. Applications of the Derivative** *4.1.* Maximum and Minimum Values *4.2.* The Mean Value Theorem *4.3.* First and Second Derivative Tests *4.4.* l'Hospital's Rule *4.5.* Curve Sketching *4.6 Optimization Problems* *4.7. Newtons Method*	HW 6, 7 Q 6, 7 Test 2 (Ch 3 and 4)
9 and 10	**Ch5. The Integral** *4.8.* Antiderivatives and *Indefinite Integrals* *5.1. The Area and Distance Problems* *5.2. The Definite Integral*	HW 8, 9 Q 8, 9

	5.3. The Fundamental Theorems of Calculus *5.4. The Substitution Method*	
11 and 12	**Ch 6. Applications of Integral** *6.1. Area Between Two Curves* *6.2. Volumes* *6.3 Volumes by Cylindrical Shells* *6.4 Average Value of a Function, Work, and Energy*	HW 10 Q10 Test 3 (Ch 5 and 6) Final Exam (All Chs)

8. Laboratory Component

Mathematica or MATLAB

9. Grading Policy

Your final grades will be determined by your results on weekly quizzes, two tests, final exam, projects, homework assignments, and class participation. The dates for each test will be announced in class one week before the test date. The materials covered in the for the m tests will be as follows:

- **Test 1** (15% of the total grade) covers Chapters 1 and 2
- **Test 2** (15%) covers Chapter 3 and 4
- **Test 3** (15%) covers Chapter 5 and 6.
- **Quizzes** (15%)
- **Homework** (10%)
- **Projects** (10%)
- **Final Exam** (20%) covers all chapters

The letter grade assignment will be determined at the end of the term.

Homework: The homework problems are posted on the course page of the SIT's LMS.
Projects: These can be found on SIT's LMS page of the course. Due dates will be announced in advance.

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For SIT's policies about Academic Integrity, Class Attendance, Classroom Behavior, Accessibility and Student Support, and Safety Considerations, please read the SIT Student Manual.

In particular the following three policies are essential for this course.

Academic Honesty: *Cheating on a test or assignment seriously undermines the integrity of the academic system and will not be tolerated. Although a student is not cheating, he or she is expected to refrain from actions that could be suspicious. Using common sense on your part should avoid unnecessary embarrassment.*

Attendance: *Attendance will be taken at every meeting. Students are expected to attend every online class and to be on time. If you are absent from any online and/or in person class for any reason, it is your responsibility to find out what you missed, including any announcements.*

Classroom Behavior: *You are expected to be in the classroom before the start of class time and leave at the end of the class time. Excessive talks without the permission of the instructors is prohibited. The use of any electronic device is allowed for the lecture and class activities only.*